

Visualise Your Delivery Pipeline in Real-Time *with Hygieia*

Hygieia is an open source project that enables the user to visualise the entire delivery pipeline. It adds support for multiple teams and a corresponding consolidated view into a program-level dashboard. Hygieia integrates widgets from story tracking, the repository, build, quality, etc, which enables easy access to important DevOps metrics.

```
$sudo update-alternatives --install /usr/bin/java java
/opt/jdk1.8.0_211/bin/java 1
$sudo update-alternatives --install /usr/bin/javac javac
/opt/jdk1.8.0_211/bin/javac 1
Select oracle-java as default one
$sudo update-alternatives --config java
$sudo update-alternatives --config javac
Check java versions
$ java -version
java version "1.8.0_211"

Java(TM) SE Runtime Environment (build 1.8.0_211-b12)
Java HotSpot(TM) 64-Bit Server VM (build 25.211-b12, mixed mode)
$ javac -version
javac 1.8.0_211
```



Hygieia

Hygieia is an open source project (released under the Apache 2 licence), which provides a single, configurable, easy-to-use dashboard for visualising the status of the entire delivery pipeline in near real-time. It provides two other self-contained dashboards — one for the DevOps engineer to track the current status of Continuous Integration Continuous Delivery(CICD) and the other for executives, to monitor the health of code commits, the deployment process and final production.

Hygieia can give complex insights into the following areas.

DevOps maturity: It offers fully automated CICD tracking of high quality and with good pipeline speed.

Risk management and investing: It connects operational metrics to developmental metrics, offering a full understanding of where to invest in order to improve processes that reduce unnecessary risks, considering the future.

Ongoing enhancements for Agile environments: It quantifies DevOps metrics to track and improve DevOps maturity.

Establishes and raises standards: It sets and achieves a profile for maturity metrics and products. When products dip below the bar, it triggers an alert notification. In addition, this bar goes up incrementally to ensure that all products undergo simultaneous improvements.

Table 1 presents an overview of Hygieia's architecture.

Table 1

Layer	Description
UI layer	The UI layer (user interface) is Hygieia's front-end and contains all the graphical user interface (GUI) elements for users to view. It is here that users are also able to configure the dashboard.
API layer	The Hygieia API layer contains Hygieia APIs and audit APIs. Hygieia APIs contain all the typical REST API services that work with the source system data (collected by service tasks) and the Internet. Hygieia audit APIs are a collection of API endpoints that serve to audit CI/CD data gathered by Hygieia collectors. This layer is an abstraction of the local data layer and the source system data layer.
DevOps tools	This layer entails the multitude of DevOps tools in a CI/CD pipeline. In Figure 1, Jira, Git, Sonar, and XLDeploy are listed as examples.
Collectors' layer	The collectors' layer fetches data from your DevOps tools. In turn, this data appears on your Hygieia dashboard. You can choose to install the collectors applicable to your DevOps tool set from the Hygieia Collectors Inventory.
Database layer	Hygieia uses MongoDB as the database for the storage and retrieval of data.